1. PREFACE

The following report details a research study undertaken in the context of Bitcoin price prediction, which implements price history information together with sentiment derived from social media platforms for Reddit. The paper also tries to address how the various social media sentiments impact on the fluctuation of price of bitcoins so that future trends in the price of bitcoins can be predicted.

Firstly, I am extremely grateful to all those who have in some way or the other helped in the completion of this research project. A very special thanks goes to my mentor for their contribution of their expertise, advice, and time during this research. In the dissection, I also want to thank the research assistants who helped in data collection, analysis, and writing the report.

The central research question guiding this study is: Can one determine and explain the degree of how sentiment on social media, particularly Reddit, is likely to affect Bitcoins? Reddit sentiment and Yahoo Finance price data are used as the major datasets for the analysis to be conducted in this study. This encompasses a range of methods for sentiment analysis, as well as the types of models that can be created. Limitations include absence of frequent data, quality of data that is sometimes obtained from social media, and issue of sentiment analysis.

This particular research seeks to establish a relationship that links social media sentiment with bitcoin so as to produce models that can be used to predict its future prices. These are as follows; To identify and summarize the time series analysis on the sentiment of Bitcoin on Reddit to establish the integration of sentiment analysis with historical Bitcoin price data Proposed an evaluation of predictive models for the Bitcoin price prediction.

The approach entails scraping data from Reddit with the help of Python and its Libraries namely PRAW which has the function of fetching data from the Reddit website and calculating sentiment of the number of posts and comments on the selected portfolio by applying different algorithms of the sentiment analysis. Cost data is obtained from Yahoo Finance. Performing data cleaning and feature engineering, exploratory analysis, and the structures of the Long Short-Term Memory (LSTM) models comprise the analytical technique.

This research attempt adds value to the existing body of knowledge by affording understanding to the research question concerning the existence of a correlation between social media sentiment and the price of Bitcoin. It is believed that the results will add to the existing knowledge base of the cryptocurrency market and potentially has implications for trading activities and investment.

This introduction outlines the flow of the subsequent sections found in this report, these include; Background to the Literature Review part where similar studies have been reviewed and compared to the current study, the Methodology part that outlines the type of research used in the study, Results that embraced the analysis of the data collected, Discussion that embraced the interpretation of the results and; Conclusion part where conclusions are drawn based on the results of the analysis.